Cybersecurity Activities

CASE STUDY 1: Ransomware Attack on a Smart Hospital

1. Detection Phase

Early Indicators:

- Sudden system lockouts across hospital devices
- Patient data becomes inaccessible
- Ransom message appears ("Pay 50 Bitcoin")
- Malfunctioning IoT devices (ventilators, monitors)
- Unusual network traffic or system slowdown

Logs to Check:

Network Logs:

- Suspicious outgoing connections
- Unknown IP communication

Device Logs:

- Unauthorized access attempts
- Firmware irregularities

Firewall Alerts:

- Blocked suspicious traffic
- Repeated intrusion attempts

2. Containment

Actions:

- Disconnect infected IoT devices from network
- Avoid shutting down all systems (critical care must continue)
- Shut down only if absolutely necessary

Immediate Steps:

- Isolate infected systems (network segmentation)
- Disable remote access
- Block malicious IPs
- Inform incident response team

3. Eradication

Possible Sources:

- Phishing email
- Unpatched IoT firmware

Removal Steps:

- Identify infected systems
- Use anti-malware tools
- Remove malicious files/processes
- Patch vulnerabilities
- Reinstall OS if needed

4. Recovery

- Restore from clean backups
- Ensure backups are not infected
- Gradually reconnect systems
- Monitor for suspicious activity

5. Forensics

Evidence:

- Network, system, firewall logs
- Malware samples
- Affected devices
- Email records

Chain of Custody:

- Ensures legal validity of evidence
- Prevents tampering
- Maintains integrity

Hands-on Task

Nmap:

- Used to scan open ports
- Helps identify vulnerabilities

Possible Entry Points:

- Open ports (e.g., SSH, RDP)
- Weak authentication
- Unpatched systems
- Phishing attack

Questions

1. Should ransom be paid?

No:

- No guarantee of data recovery
- Encourages cybercrime

- Risk of repeated attacks

2. How IoT increases attack surface?

- More connected devices
- Weak security in IoT
- Lack of updates
- Poor authentication

3. Preventive Policies:

- Regular updates and patching
- Multi-factor authentication
- Network segmentation
- Employee training
- Regular backups
- IoT security standards

CASE STUDY 2: Data Breach in Cloud-Based E-Commerce

1. Root Cause

- Misconfigured cloud storage (public access)
- Weak authentication

- Lack of monitoring

2. Incident Response Lifecycle

Detection:

- Data found on dark web
- Security alerts
- Suspicious access logs

Containment:

- Disable public access
- Restrict permissions
- Block unauthorized users

Eradication:

- Fix configurations
- Strengthen access control
- Patch vulnerabilities

Recovery:

- Notify users
- Reset passwords
- Restore secure systems

- Rebuild trust

3. Cloud Security Challenges

Shared Responsibility Model:
- Provider handles infrastructure
- Company handles data and access

- Data encryption issues
- Identity and access mismanagement

Questions

1. Who is responsible?
Both, but mainly the company:
- Provider handles infrastructure security
- Company handles data protection and configuration

2. Legal Implications:
- Violation of data protection laws
- Fines and penalties
- Customer lawsuits
- Reputation damage

3. Securing Cloud Infrastructure:

- Strong IAM policies
- Data encryption (at rest and in transit)
- Regular audits
- Firewalls and intrusion detection systems
- Disable public access
- Multi-factor authentication